William Linch

Director of Astronomy and Astrophysics

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Teaching and service:

- Advanced Astronomy: Solar System (fall)
- Advanced Astronomy: Universe (spring)
- Astronomy Research Lab
- Outreach

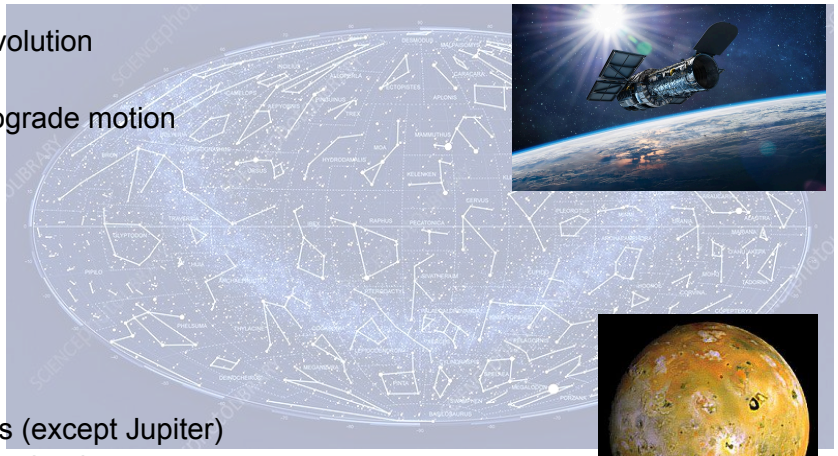
Background:

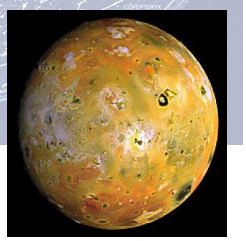
- Ph.D. in theoretical physics from UMD (2005)
- Research positions and professorships
    - University of New York at Stony Brook
    - Universidad Andres Bello in Santiago, Chile
    - University of Maryland at College Park
    - Texas A&M University
- Started at TJ in 2023

Fun fact:

- College roommate and friend of Dr. Osborne (across hall)

Advanced Astronomy: Solar System

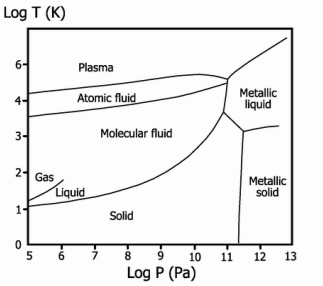


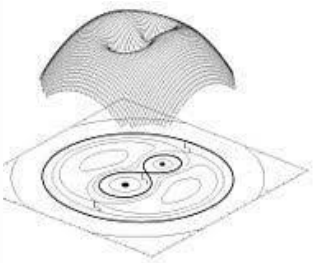


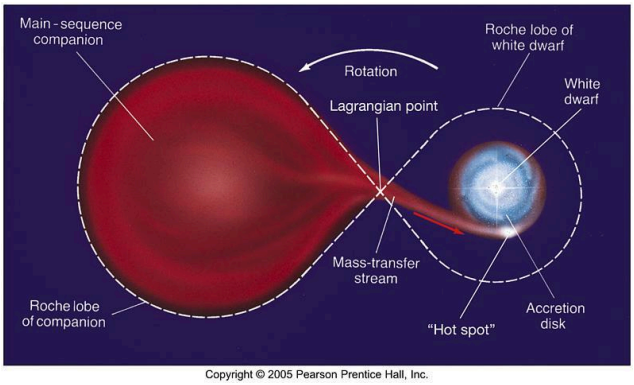
- Aristarchus and the Copernican Revolution
  - Solstices and equinoxes
  - Star charts, the ecliptic, retrograde motion
  - Heliocentrism
- Telescopes
- Astrodynamics
- Formation of the solar system
- Astrophysics of the Sun
- Astrophysics of Jupiter
- Projects on (time permitting)
  - Terrestrial and Jovian planets (except Jupiter)
  - Asteroid belt/Kuiper belt/Oort cloud

In the process, we will learn astrophysics concepts and how to apply them including

- Astro-geometry
  - Parallax
  - Angular sizes and distances
  - Ellipses and eccentricity
- Newtonian astrophysics
  - Kepler's laws
  - Mass determination and barycenter
  - Orbital and escape speeds
  - Atmospheres
  - Rotating frames of reference, Lagrange points, and orbital dynamics
  - Roche limits and fragmentation
- Nuclear and Particle physics I
  - Simplified Model of Particle Physics
  - Mass-energy equivalence
  - Nuclear reactions







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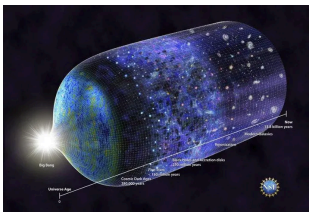
Advanced Astronomy: Universe

- Solar neighborhood and exoplanets
- Stellar formation and evolution
- Star death:
  - Novea
  - Supernovae
  - Neutron stars
  - Black Holes
- Galaxies
  - Dark matter
- Cosmology
  - Big bang
  - Dark energy



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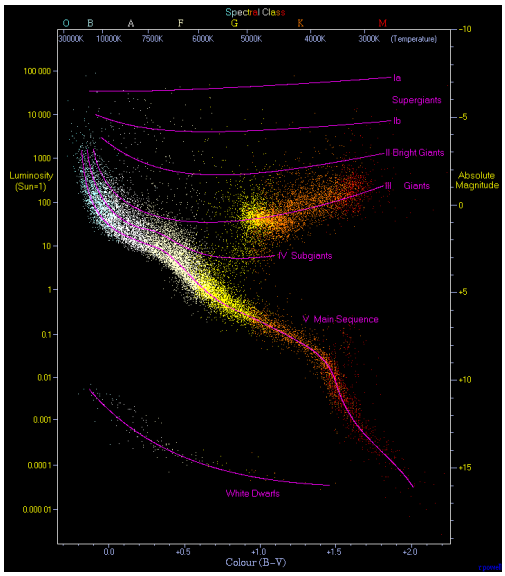
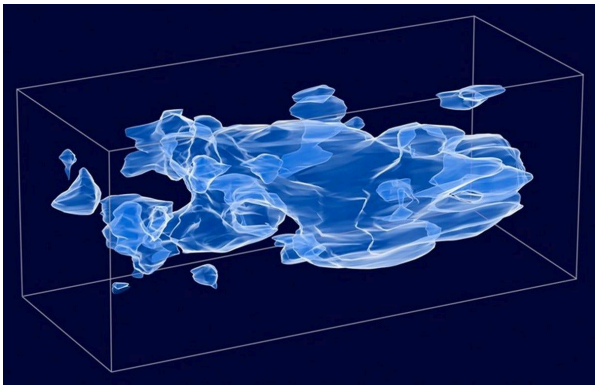
- Waves
  - Doppler effect
  - Spectrum
  - Blackbody radiation
- Quantum theory: Bohr model of atomic structure
- Spectroscopy: The interaction of atoms and radiationAstrophysics of multi-body systems
- Special and general relativity
  - Time dilation and length contraction
  - Event horizons
- Nuclear and particle physics
  - (Beyond?) the Standard Model of Particle Physics
  - Nuclear fusion
  - Electron degeneracy and the Chandrasekar mass
  - Neutron degeneracy
- Nuclear and Particle physics II
  - The Standard Model of Particle Physics